

Indolent and mantle B-cell lymphoma

Follicular and **mantle cell** — grouped because both are managed as chronic disease with BTK inhibitors, bendamustine and maintenance, not because mantle cell is indolent. **It is not:** it is biologically aggressive and behaves like a disease you manage rather than cure

Stanford Oncology Handbook 2025, Lymphoma: Non-Hodgkin, pp.223–232 · NCCN B-Cell Lymphoma v2.2025 · items marked † are outside the handbook chapter

FOLLICULAR GRADE

Grade	Incidence	Histology and behavior
I	40–45%	0–5 centroblasts/hpf — indolent, "classic FL" (WHO 2022)
II	30%	6–15 centroblasts/hpf — indolent, classic FL
IIla	20%	>15 centroblasts/hpf, centrocytes present — indolent or aggressive
IIlb	—	>15 centroblasts/hpf, centrocytes absent , sheets of centroblasts — aggressive, behaves like DLBCL

FLIPI — “NOLASH”, 1 POINT EACH

Risk	%	5-yr OS	10-yr OS
Low (0–1)	36	91%	71%
Intermediate (2)	37	78%	51%
High (3–5)	27	53%	36%

Nodal sites >4 · LDH elevated · Age ≥60 · Stage III–IV · Hgb <12. FLIPI2 instead uses age >60, β2-microglobulin > ULN, marrow involvement, Hgb <12 and largest node >6 cm — 5-yr PFS 80% / 51% / 19%.

GELF — WHEN TO TREAT FOLLICULAR

- ≥3 nodal areas each ≥3 cm
- Any nodal or extranodal mass ≥7 cm
- Symptoms attributable to disease, including B symptoms or effusions/ascites
- **Cytopenias** from lymphoma (WBC <1,000 and/or platelets <100,000)
- Leukemic phase >5,000 cells/μL
- Splenomegaly

Meeting none of these is a reason to observe, not to treat. Brice, JCO 1997 · NCCN v2.2025.

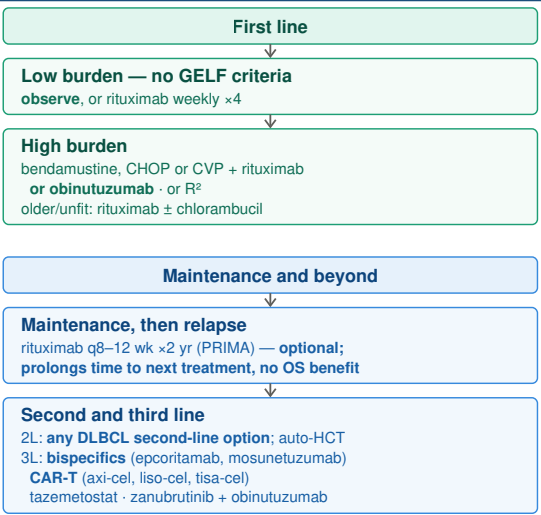
MIPI — MANTLE CELL

Points	Age	ECOG	LDH / ULN	WBC K/uL
0	<50	0–1	<0.67	<6.7
1	50–59	—	0.67–0.99	6.7–9.99
2	60–69	2–4	1.00–1.49	10–14.99
3	≥70	—	≥1.50	≥15

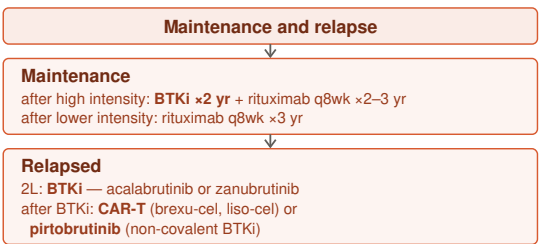
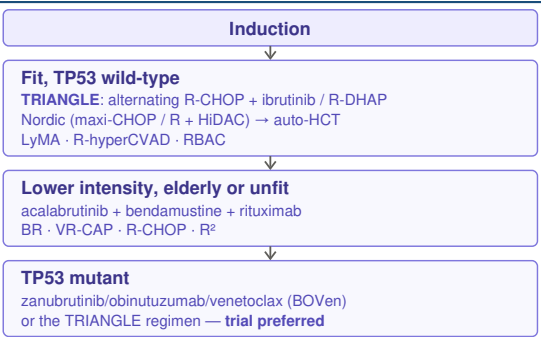
Hoster, Blood 2008. **The handbook prints the points table but not the risk-group cutoffs or their survival.**

TP53 in mantle cell. Mutated in ~10% and a strong predictor of poor outcome with chemotherapy **and** with autologous transplant, and also with ibrutinib. **Send NGS at diagnosis;** prefer a clinical trial. Deep responses have been seen with targeted combinations — **BOVen** (zanubrutinib/obinutuzumab/venetoclax) and OAsIs.

FOLLICULAR PATHWAY



MANTLE CELL PATHWAY



The case for autologous transplant in mantle cell has collapsed. The old PFS benefit predates BTKi and maintenance. TRIANGLE found no benefit once ibrutinib was added, and EA4151 none for patients in MRD-negative CR.

REGIMENS

Regimen	Dosing	Cycle
Rituximab alone	375 d1, 8, 15, 22	single course
R-bendamustine	Rituximab 375 d1; bendamustine 90 d1+2	q28d ×6
R-CHOP · R-CVP	Rituximab 375 · cyclophosphamide 750 · vincristine 1.4 (max 2 mg) d1 + prednisone 40/day d1–5, with doxorubicin 50 (CHOP) or without (CVP)	q21d ×6–8
Maintenance rituximab	375 q2 months ×2 years (PRIMA). In MCL: 375 weekly ×4 q6 mo ×2 yr, or q3 mo ×2 yr	—
Obinutuzumab	1000 mg d1, 8, 15, then monthly C2–6/8, then bimonthly ×2 yr	—
R ²	Rituximab 375 d1; lenalidomide 20 mg PO d1–21 + ASA 81 mg	q28d ×12
R-GemOx	Rituximab 375 · gemcitabine 1000 · oxaliplatin 100, all d1	q14d ×6–8
PEP-C (palliative, oral)	Prednisone 20 + cyclophosphamide 50 + etoposide 50 + procarbazine 50 mg, all PO daily until WBC <3,000, then resume on a reduced schedule	continuous
Mod. R-hyperCVAD	Rituximab 375 d1; cyclophosphamide 300 q12h d1–3 (6 doses); doxorubicin 25 CIV d1+2; vincristine 2 mg d3; dexamethasone 40 mg d1–4	q28d ×6, then maintenance R
R-DHAP	Rituximab 375 d1; cisplatin 100 CIV d1; cytarabine 2000 q12h ×2 d2; dexamethasone 40 mg d1–4	q21d ×3
VR-CAP	Bortezomib 1.3 SQ d2, 5, 9, 12; rituximab 375 + cyclophosphamide 1000 d1; prednisone 100 mg d2–6	q21d ×8
BTK inhibitors	Ibrutinib 560 mg daily · acalabrutinib 100 mg BID · zanubrutinib 160 mg BID (or 320 mg daily) — continuous until progression	continuous

mg/m² unless stated. Cytarabine → 1 g/m² if age >60, with prednisolone 1% eye drops. R-GemOx needs ANC ≥1,000 and platelets ≥100,000 to proceed. Consider aspirin with lenalidomide.

PEARLS

- **Follicular grade IIlb is not an indolent lymphoma** — centrocytes are absent and it is treated as DLBCL
- **Maintenance rituximab buys time to next treatment, not survival** — worth saying out loud when consenting
- **Mantle cell TP53 mutation changes the plan**, not just the prognosis — chemotherapy, transplant and ibrutinib all underperform; go to trial or a targeted combination
- † The handbook covers no mantle cell biology — **cyclin D1, t(11;14), SOX11, Ki-67 and the blastoid variant are absent**, as is the indolent leukemic-non-nodal subset that can be observed

TRIALS

Trial	Role
PRIMA	Maintenance rituximab ×2 yr after frontline immunochemotherapy in follicular
StiL NHL1	BR vs R-CHOP — the basis for BR frontline
GALLIUM	Obinutuzumab vs rituximab with chemotherapy in follicular
RELEVANCE	R ² vs R-chemo — negative superiority trial ; R ² is equivalent
TRIANGLE	Ibrutinib in induction + maintenance — no benefit from auto-HCT (endpoint: FFS)
ECOG-ACRIN EA4151	No auto-HCT benefit in MRD-negative complete remission
SHINE · ECHO	BTK inhibitor added to chemoimmunotherapy in older MCL

Numbers on the Hub cards come from the primary publications — the handbook prints citations without results.